



Corbett, Greville G. 2026. The typology of government by adpositions: Towards an understanding of “differential”. *Glossa: a journal of general linguistics* 11(1). pp. 1–33. DOI: <https://doi.org/10.16995/glossa.24698>



The typology of government by adpositions: Towards an understanding of “differential”

Greville G. Corbett, Surrey Morphology Group, University of Surrey, Guildford, UK; Department of Human Behavior, Ecology and Culture, Max Planck Institute for Evolutionary Anthropology, Leipzig, Germany, g.corbett@surrey.ac.uk

Adpositions may govern case, and there is dramatic variation in the case values found. Different adpositions often govern different cases, which means that we need to partition (or split) the adpositions according to the case(s) they take, as in German. But this is just one possibility. The typological point of interest here is that the items governed (the governees) may also be a determining factor, as when nouns and pronouns require different cases, in Mari for instance. Furthermore, we may need to specify both the adposition and the governee (Russian). And then, rather than one fixed outcome, there may be alternating outcomes, for each of the three types of partition (adposition-based, governee-based, and splits based on both). At the extreme, even when we have specified a particular adposition, and an individual noun, we may still need further conditions. Here corpus work reveals the interesting interaction of determining factors, notably in Finnish. Our exploratory canonical typology lays out the possibilities, providing clear differentiation of the attested systems and subsystems. The specific example of adpositions can then be used as an entry point to approach unresolved issues in the more general area of differential marking.

Glossa: a journal of general linguistics is a peer-reviewed open access journal published by the Open Library of Humanities. © 2026 The Author(s). This is an open-access article distributed under the terms of the Creative Commons Attribution 4.0 International License (CC-BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See <http://creativecommons.org/licenses/by/4.0/>. Articles published by the Open Library of Humanities may also contain other CC-licensed material, or material used or reproduced under an exception or limitation to copyright. Such material is not subject to the article's license. It is either marked with its own license, or marked that it has been reproduced with permission or under fair-use.

OPEN ACCESS



1 Introduction

Splits in case government, as when a given adposition may govern accusative or dative, are pervasive. Our theory of such splits is patchy. I therefore propose an initial typology, to open the way for a fuller understanding of the phenomenon, from minor splits eventually to differential argument marking. As a manageable but illuminating pilot study, I focus on case as governed by adpositions. Specifically, I take the approach of Canonical Typology, in exploratory mode (Bond 2013: 24–25) to specify the theoretical possibilities of splits in adpositional government and then establish which are found. Given the detailed evidence required, it is natural that better-studied languages figure large, including Russian, German and Finnish. But where researchers have paid particular attention to relevant data, other languages offer vital evidence too, as we shall see with Armenian and Mari.¹

A well-grounded analysis of case (Zaliznjak 1973; Goddard 1982; Corbett 2012: 200–222) “should” predict the appropriate case value in any circumstance. A reasonable starting point is that a single case value will be specified. In adopting this as a baseline, I follow the standard scientific method of measuring from the simplest instances of each dimension of variation (Round & Corbett 2020). This simplest instance is that in which all adpositions behave alike, as do all governees (§2). This proves a fine baseline which allows us to examine, in a principled way, the systems which are more complex. First, the instances where we need to partition the adpositions (different adpositions govern different cases), then those where we need to partition the governees (different governees require different cases), and then those where we need to partition both adpositions and governees (§3). In the first instances analysed we are dealing with one result. We then move on in §4 to those where there are alternative possibilities, for each of the three types (adposition-based, governee-based, and splits based on both), making it important to specify the conditions on the choice (§5).

Adpositions turn out to be fascinating in their own right. And there are clear advantages in looking just at adpositions and their government properties, since they form a manageable subclass of problems within larger systems of case. By taking them as our focus, we can factor out some of the variables. They allow us to draw attention to the prevalence of morphosyntactic splits. But going further, they also allow us to make progress on the place of differential marking within the typology of marking (§6).

¹ We need the careful groundwork on the analytical distinctions, as a pre-requisite for large-scale surveys (Bonmann & Riesberg & Himmelmann 2025: 158). And exploratory typologies can form the basis for in-depth studies of particular languages and families (Micali & Vincent & Wheeler 2025).

2 Canonical interaction of adposition and governee

To ground our typology, we need a clearly definable (and clearly recognizable) baseline. Here is such a proposal:

In a canonical system of adpositional government, all adpositions govern the same case value on every governee (complement).

This is the logical baseline to calibrate from. We aim to clarify the conflicting factors which have hampered research, by isolating the different ways in which real examples deviate from this artificial ideal. Such a baseline system can be represented as in **Table 1**.

	governee₁	governee₂	governee₃
adposition ₁	A	A	A
adposition ₂	A	A	A
adposition ₃	A	A	A

Table 1: Canonical system of adpositional government.

Note: ‘A’ is a value of case.

Table 1 includes three adpositions and three governees (of whatever type); provided they are representative, the schema scales up to larger real systems and subsystems. According to our baseline definition, all the adpositions (of which three instances are given) behave identically in the canonical instance. They govern the same value (“A” in **Table 1**) of the feature case.² This holds irrespective of the way in which the case values are realized, whether through different affixes, alternations or the bare stem. Moreover, in the canonical world, the possible governees (of which three are given) behave identically in terms of the value governed (Corbett 2012: 193–194). We require that for a given language we have established a coherent account of: (i) the items which count as adpositions, and (ii) the case system.

- (i) To establish the **adpositions**, we may use general criteria as in Svenonius (2007: 67–77), Dryer (2013), Libert (2013) and Herce (2023) among others.³ For surveys see Hagège (2010) and Libert (2016).

² This holds for government more generally, including verbal government; our focus is more narrowly on adpositional government.

³ There are also interesting additional language-specific criteria (for instance, certain Russian pronouns take an initial *n*- after prepositions, Hill 1977, while Gouskova 2019: 185–201 provides further tests).

This definition of adposition offers a good starting point:

a class of words (i.e. not bound formatives like case markers, but rather independent words) that take a noun or noun phrase as their complement and specify its grammatical or semantic relation to another element, typically a verb (e.g. providing information about the time or location of the event) but also a noun (e.g. in possessive constructions) in the same clause. (Herce 2023: 420)

And after demonstrating the need to broaden the traditional definition of preposition, Huddleston & Pullum (2002: 603) offer this definition:

a relatively closed grammatically distinct class of words whose most central members characteristically express spatial relations or serve to mark various syntactic functions and semantic roles.

- (ii) For establishing the **case system** (independently of adpositions), seminal work was conducted within the Set-theoretical School, for which see particularly Zaliznjak (1973); Goddard (1982) is an important contribution, and the key issues are summarized in Corbett (2012: 200–222).

This definition provides a good starting point:

Case is a system of marking dependent nouns for the type of relationship they bear to their heads. Traditionally the term refers to inflectional marking, and, typically, case marks the relation of a noun to a verb at the clause level or of a noun to a preposition, postposition or another noun at the phrase level. (Blake 1994: 1)

In both instances we shall aim for the clearest examples (canonical adpositions and canonical case values). In this way, we sidestep the “coast fallacy”, according to which the coast exists as an indeterminate area, which we cannot securely define as either land or sea, therefore we cannot use the constructs ‘land’ or ‘sea’. We progress by the common-sense route of using clear-cut examples of adpositions (‘words’ in both definitions) and case values (‘inflectional marking’); this is equivalent to taking Berlin as our example of land, and the middle of the Atlantic as our representative of sea.

Since in our simplest system the adpositions and the governees show no variety, **Table 1** boils down to **Table 2**:

	governees
adpositions	A

Table 2: The simplest system of adpositional government.

The system in **Table 2** provides an excellent baseline to measure from. In Canonical Typology we specify such baselines, to ensure comparability between accounts (Round & Corbett 2020). It is not a problem if there are no canonical examples (any more than the dearth of objects with a temperature of zero kelvin is a problem for physicists).

In this instance, however, we do find real examples. In North Saami adpositions take the genitive (Grünthal 2019: 12); in Yiddish all prepositions take the dative, according to Libert (2002: 135), who gives other possible example languages. The limiting instance of ‘all’ is where there is only one adposition, hence all behave alike; Dryer (2013) suggests this is the case in Wardaman (Yangmanic, spoken in the Northern Territory of Australia) based on Merlan (1994: 99). And there are languages outside our scope. First, those which appear not to have adpositions (Dryer 2013); Foley (2023: 131), citing Scorza (1985), suggests the Torricelli language Au as an example. And second, those without morphological case, like Vietnamese (Ngo 2021).

3 Categorical splits in the interaction of adposition and governee

In the canonical situation shown in **Tables 1** and **2**, all adpositions govern the same case value of every governee.⁴ This situation can be found, but it appears to be rare. We find numerous departures from it: in particular, we find additional feature values available (as we see, for instance, in **Table 5**). To account for their distribution, we may need to partition the adpositions (§3.1), or the governees (§3.2), or both (§3.3). Such an extra partition is often termed a split (Corbett 2023a: 113). We look at the range of possibilities, from free combinations to fixed phrases (§3.4), and then pause to consider some underlying assumptions, based on the examples examined so far (§3.5).

3.1 Partitioning the adpositions

By setting up a part of speech ‘adposition’ for a given language, we imply that we are grouping items with shared syntactic behaviour. In the canonical world, members of such a part of speech would have consistent syntactic behaviour (for instance, all would be prepositions or all postpositions); often, however, their behaviour is not consistent. We concentrate on government, and find that we need to split the adpositions according to the case which they govern.⁵ We may diagram the situation as in **Table 3**.

⁴ Compare Corbett (2023a: 112): ‘a baseline lexeme always requires the same specification of other lexemes (it is the same for others); if a postposition requires the genitive, it does so for all items it governs’.

⁵ There may be other reasons to split the adpositions, of course; thus some adpositions may be agreement targets while others are not (Corbett 2006: 46, Bond & Chumakina & Kaye 2023: 8–10). Herce (2023) shows clearly how far adpositions cluster together in their different behaviours in Basque, German and Hungarian. For classification of adpositions compared with case values, which can be tricky in some languages, see Hentschel (2003).

	governee₁	governee₂
adposition ₁	A	A
adposition ₂	B	B

Table 3: Split adpositions (schematic).

This is a relatively familiar situation. Consider these simple phrases:⁶

- (1) *Russian* (rus)
 okolo škol-y
 near school-SG.**GEN**
 ‘near the school’
- (2) za škol-oj
 behind school-SG.**INS**
 ‘behind the school’

While otherwise syntactically similar, these phrases differ in the case governed by the prepositions: genitive in (1) and instrumental in (2). This instantiates our schema (**Table 3**) as shown in **Table 4**.

	governee₁	governee₂
<i>okolo</i> ‘near’	GENITIVE	GENITIVE
<i>za</i> ‘behind’	INSTRUMENTAL	INSTRUMENTAL

Table 4: Split adpositions (Russian).

Finding just two adpositions which behave differently is sufficient to demonstrate the split. However, this subsystem fits within a larger system, showing multiple instances of this same type of split. This is clear from **Table 5**, based on Wade (1992: 416–417), in turn starting from Wade (1983). The table includes just what Wade calls the ‘primary’ prepositions. For each I give the case values governed, with an indication of the meaning for the different uses.

⁶ I follow the Leipzig Glossing Rules, and I use bold as a flag to draw attention to relevant characteristics of examples. Abbreviations are the Leipzig ones with additions (see list at end of paper). Languages are identified by their ISO 639-3 code.

preposition	case governed and gloss				
	ACCUSATIVE	GENITIVE	DATIVE	INSTRUMENTAL	LOCATIVE
čerez	‘across’				
pro	‘about’				
v	‘in’ (motion)				‘in’
na	‘on’ (motion)				‘on’
o	‘against’				‘about’
pod	‘under’ (motion)			‘under’	
za	‘behind’ (motion)			‘behind’	
po	‘up to’		‘along’		‘after’
s	‘approximately’	‘down from’		‘with’	
bez		‘without’			
dlja		‘for’			
do		‘as far as’			
iz		‘out of’			
krome		‘except for’			
ot		‘from’			
radi		‘for the sake of’			
u		‘at’			
k			‘towards’		
meždu				‘between’	
nad				‘above’	
pered				‘in front of’	
pri					‘in the presence of’

Table 5: Government possibilities of Russian primary prepositions.

Table 5 demonstrates clearly the need to partition the Russian prepositions. They can govern not only the genitive (1) and instrumental (2) but also the accusative, dative and locative.⁷ At the most basic level there is clear split, since prepositions govern differently. (See Timberlake 2004: 174–184 for a clear summary of the system.) That is the point of **Table 5**, for the present. We also see that several prepositions show a split in that they have more than one possibility; for instance, some distinguish location (place) versus motion (path), a common characteristic to which we return in §4.1, where we discuss more complex splits of adpositions.

Governees are not given in **Table 5**, since in almost all instances they are not relevant; the case value to be used depends on the preposition (there are few exceptions, see §3.3. for discussion of the second locative, while the remarkable role of the governee with *po* ‘up to, along, after’ is referenced in the discussion of Finnish in §4.3). Other languages, however, give the governee a much greater role, and we now turn to examples of that.

3.2 Partitioning the governees

There may also be splits in the other dimension; that is, case requirements may differ according to the governee, as diagrammed in **Table 6**.

	governee ₁	governee ₂
adposition ₁	A	B
adposition ₂	A	B

Table 6: Split governees (schematic).

This type of split is shown by data from Mari (chm), formerly known as Cheremis (a Uralic language spoken mainly in the Mari El Republic of the Russian Federation), where the governees of postpositions are split. Personal pronouns stand in the genitive, as in (3).

- (3) *Mari* (in the Uralic Phonetic Alphabet, Riese & Bradley & Yefremova 2022: 156)
memnan ončəlno
 1PL.GEN in.front.of
 ‘in front of us’

⁷ There is an apparent instance of a Russian preposition governing the nominative (the *idti v letčiki* ‘become a pilot’ construction); for discussion, and an alternative analysis avoiding government of the nominative here, see Corbett (2012: 210–213), following Mel’čuk (1985: 461–489).

Lithuanian provides an interesting comparison to Russian. According to Ambrazas et al. (1997: 406–407, 420–421) most prepositions take one case value (the most popular is genitive, then accusative, then instrumental). In addition, *už* ‘behind, over, outside; later; by, for, etc.’, takes genitive or accusative (p. 420), as do a handful of others (p. 407); *po* ‘about, around, after’ uniquely has three possibilities, namely genitive, accusative or instrumental.

However, nouns (and extended nominal complements) normally appear in the nominative (which has no overt marker in (4)):

- (4) *Mari* (Riese & Bradley & Yefremova 2022: 156)
 škol ončâlno
 school[SG.NOM] in.front.of
 ‘in front of the school’

For further examples see Alhoniemi (1993: 50), Kangasmaa-Minn (1998: 236–237) and Burukina (2023); the major source is Riese & Bradley & Yefremova (2022), who focus on Meadow Mari (mhr). These examples show the need for additional partitioning of the governees. While we might expect that governees would behave alike, here they are split, as shown in **Table 7**.

	personal pronoun	noun
adposition ₁	GENITIVE	NOMINATIVE
adposition ₂	GENITIVE	NOMINATIVE

Table 7: Split governees (Mari).

The account given is true in broad-brush terms. There is a good deal of complicating detail, for which see especially the careful account in Riese & Bradley & Yefremova (2022: 154–191). In particular, a minority of adpositions govern other case values (exclusively or in addition to the usual pattern); this includes the possibility of taking personal pronouns in the nominative in some persons (Riese & Bradley & Yefremova 2022: 157), a point taken up in §3.5. There are complex patterns of combinations of pronouns and adpositions, including instances where adpositions border on being case markers. Hence **Table 7** presents an idealization of a part of the system of Mari.

More generally, governees can be split in two different ways. We may find a **lexemic** split when different governees are governed differently, as in Mari and in Armenian (to be analysed in §4.2). But we may also find a **featural** split of the governees, where the governee of an adposition stands in a different case according to its own number value; see Corbett (2023a: 126–127) for an example from Belarusian.⁸ For present purposes we will focus on lexemic splits of governees.

⁸ The remarkable system of the Baltic language Latvian (lav) might appear to fit here. Prepositions govern different case values, according to the governee’s number: prepositions govern various case values in the singular (accusative, genitive or dative), but they all take the dative in the plural (Corbett 2023b: 5–6, with key references there). The solution I have proposed is to augment the case feature with two new values, mnemonically accusative-dative and genitive-dative. Prepositions then govern a particular case value (there is no split) and the relevant syncretisms are realized by the inflectional morphology. This is formalized in Corbett (2010). Partitioning the prepositions would not work, since the behaviour is fully general for prepositions (unlike the Mari system). Partitioning the governees would not work, since the rule of government is fully general. Hence augmenting the feature system makes sense, since doing so gives us a system which is regular and categorical. For listing vs general rules see Noyer (2005: 273) and Brown & Hipsley (2012: 46).

3.3 Partitioning of adpositions and governees

Our two major types of split (in adpositions and in governees) may interact. That is, some adpositions govern differently in a way that targets just some governees. We can represent this situation as in **Table 8**.

	governee ₁	governee ₂
adposition ₁	A	A
adposition ₂	A	B

Table 8: Split adpositions and split governees (schematic).

We see an instance in Russian. Some prepositions govern the locative, and within those, there are two (*v* ‘in’ and *na* ‘on’) which in locational use govern the “second locative”. However, only some nouns have a dedicated second locative (other nouns use the normal locative). This situation is worth setting out step by step. First, we see the preposition *o* ‘about, concerning’ which in this meaning simply governs the locative case, for all types of noun:

- (5) *Russian*
- a. *o* *gazel-e*
 about newspaper-SG.LOC
 ‘about the newspaper’
 - b. *o* *sad-e*
 about garden-SG.LOC
 ‘about the garden’

In each the sense is ‘about, concerning’ and the locative case is found. Contrast the situation with *v* ‘in’:

- (6)
- a. *v* *gazel-e*
 in newspaper-SG.LOC
 ‘in the newspaper’
 - b. *v* ***sad-u***
 in garden-SG.LOC2
 ‘in the garden’

This preposition can govern the second locative, when available. With most nouns, *v* ‘in’ takes the locative, as in (6a); a listable minority of nouns have a second locative, as illustrated by *sad* ‘garden’ in (6b); for more examples and analysis see Brown (2007) and Corbett (2012: 206–209); for the semantics see Plungian (2002), and for the possible role of contact with Finno-Ugric languages in the development of the second locative see Breu (1994: 50–52).

We can represent this situation as in **Table 9**.

	<i>gazeta</i> ‘newspaper’	<i>sad</i> ‘garden’
<i>o</i> ‘about’	LOCATIVE	LOCATIVE
<i>v</i> ‘in’	LOCATIVE	SECOND LOCATIVE

Table 9: Split adpositions and split governees (Russian).

Both the adpositions and the governees are split.⁹ At first glance, this is less of a departure from the canonical situation of **Table 1** than either **Table 3** or **Table 4**. But it is more serious, since it requires two splits rather than one. Moreover, this split is embedded in the complex Russian system given above in **Table 5**.

3.4 From free combination to fixed phrase

The sequence from §3.1 to §3.3 is not accidental. We first considered splits in the adposition, where different adpositions govern different case values (§3.1). This may well involve straightforward instances of free combinations, where adpositions take a wide range of governees, and impose their government requirement on them consistently. When we move to the splits in governees (§3.2), we are already leaving the domain of free combinations; the adposition needs to “peek” at the governee to determine which case it will govern. It is significant that the examples here often involve a split between pronouns and nouns, as occurs in other instances of case marking split.¹⁰ And the third type, involving splits in both the adpositions and the governees (§3.3), suggests phrases which are at least partially fixed. Thus the Russian second locative (§3.3) is restricted to the prepositions *v* ‘in’ and *na* ‘on’; but a given noun typically occurs with only one of these: *v sadu* ‘in the garden’ but *na mostu* ‘on the bridge’. The noun determines the preposition, and the preposition governs the special second locative.

Such restrictions on adpositional phrases (from both adposition and governee), which are signs of partially fixed expressions, can be manifested in various ways besides those which are our focus. There can be restrictions on items intervening between adposition and governee. Their meaning can become less than fully compositional. There may also be rather specific rules, as for instance with the Latin preposition *cum* ‘with’, which is postpositional just with a small inventory of items, including the personal pronoun, as in *mecum* ‘with me’ (see Fortson 2010: 136–140 and references there).¹¹ And in Russian, the preposition *o* ‘about, concerning’ (taking the locative,

⁹ There is a further split in the governees in that the second locative is found only in the singular.

¹⁰ The split need not run along part of speech lines: Turkish shows a split within the pronouns (Corbett 2013: 128–129), and references there; this is also true of Armenian (§4.2).

¹¹ A further indicator of the way in which adposition plus governee may move towards being a fixed phrase is found when the case system is being lost. We might expect that case would be lost first in adpositional phrases, since in

Table 5) has this as its basic form; there is a shape condition such that before vowels it has the form *ob* (but with some exceptions both ways). In the specific phrases *obo mne* ‘about me’ and *obo vsem* ‘about everything’ it has the third form *obo* (Timberlake 2004: 179); also before other forms of *ves* ‘all, every’, such as *obo vsech delax* ‘about all matters’.¹² At the extremes, there are even situations where it is arguable whether we have adpositions inflected for person and number or personal pronouns with adpositions realizing case; see Stewart & Joseph (2009) and Hutchinson (2025) on Scottish Gaelic, and Lamb (2025: 159–192 for a wealth of examples), and Spencer & Stump (2013) on Hungarian.

3.5 Assumptions

I take it as given that the existence of languages like Mari, where familiar rules of government are not fully adequate, are of interest, both for syntax and for typology. And this is true whether such languages are few or numerous. However, a referee suggests that splits in the governee can always be handled by enriching the feature system (introducing an extra case value to be realized differently on different governees), and so the issue is not of general interest. I believe that this reasoning misses the point in two different ways: (i) it would not remove the interest; and (ii) it does not work.

Suppose first that we could find a solution for each language presented, in which by enriching the feature system we could “handle” apparent instances of governee splits, still using a single rule of case government. Even then it would remain true that English and Mari are not equivalent. Somewhere in the grammar we need to acknowledge that there is an interesting difference, which goes beyond the need for an additional case value.

And second, the proposal fails. This can be seen most readily as soon as a system is not fully categorical. We noted in §3.2 that there are complications in Mari, including the possibility of a minority of adpositions governing personal pronouns in the nominative in some persons (Riese & Bradley & Yefremova 2022: 157). This complication is helpful in that it rules out the suggestion of introducing an adpositional case, which would be realized as the nominative for nouns and the genitive for pronouns. This idea would give a straightforward syntactic rule of adpositional government, reserving the complications for the inflectional morphology. However, since there are instances where adpositions can govern pronouns in the nominative, we would be no further on, since adpositions could then govern pronouns in the adpositional (= genitive

many instances the syntactic role of the phrase is clear from the adposition alone. However, a study of Bulgarian and Serbian dialects reveals that there the use of the preposition favours retention of the original case marking (Krasovitsky & Kyuseva & Baerman & Corbett 2026).

¹² Even given its limited range, *obo* is still frequent. This is seen in Ljaševskaja & Šarov (2009: 1046), who produce counts based on 92 million words from the Russian National Corpus, and give the three forms separately. *O* is ranked 31st overall, *ob* is ranked 129th, while the rank of *obo* is 1958.

for pronouns) or the nominative. Thus introducing an adpositional case value for Mari would be an unhelpful complication.

The situation in Russian looks more promising for a simple featural solution. In §3.3 I focused on the clearest instances. With these, since the second locative has a unique form for some nouns, we could treat it as an additional case value, governed by *v* ‘in’ and *na* ‘on’ in their locative use. All governed nouns would take this case value, but it would be realized distinctly only for those nouns with a unique second locative form (and otherwise by the normal locative). However, this additional value of the feature case is not all that is needed; in some instances, there is a gradient effect, of the type to be discussed in the next section. Specifically, there are some nouns which have a second locative but do not always use it. Instead, it alternates with the normal locative, even with the prepositions *v* ‘in’ and *na* ‘on’ in locational use. This complication is significant. For those nouns which allow but do not require the second locative with *v* ‘in’ and *na* ‘on’, we need to recognize an opposition between locative and second locative, and its resolution depends on the specific noun which is the governee. Since gradient effect prove important in the choice of possible analyses, we turn to them next.

4 Gradient splits in the interaction of adposition and governee

In the instances so far, our primary focus has been on single outcomes. In the canonical instance there is a single deterministic outcome (which is equivalent to zero on a measurement scale). But there are situations where, even when both the adposition and the governee are held constant, we find more than one outcome. This is a less canonical situation:

Canonically: Single outcome > more than one outcome

(Here “>” indicates ‘more canonical than’.) ‘More than one outcome’ covers a range of possibilities. First, the outcomes can be conditioned by factors of different sorts (and by combinations of such factors), as will be discussed in §5. Second, the choice of outcome may be categorical or gradient. There has been some confusion here (see Witzlack-Makarevich & Seržant 2018: 27–29, and Corbett 2023a: 113, 119–120 for discussion); it therefore makes sense not to muddy the waters by giving single labels (such as ‘fluid’) to particular combinations. Rather we should: (i) be clear that all such instances are splits; (ii) give the conditioning factor(s); and (iii) specify whether the split is categorical or gradient, that is, whether the conditioning factor(s) uniquely specify one outcome, or not. We keep these points in mind as we now tackle instances where specifying the governor (adposition) and the governee is not sufficient to determine the case value, since there remains an alternation of case values.

4.1 Splits of adpositions with alternation of case values

We start with the splits triggered by a particular adposition or group of adpositions (shown in Table 10).

	governee₁	governee₂	governee₃
adposition ₁	A	A	A
adposition ₂	A/B	A/B	A/B
adposition ₃	B	B	B

Table 10: Split adpositions: plus alternation of case values (schematic).

Note that A/B in our tables indicates the possibility of two different case values, each of which is justified by independent analysis, and each of which can occur with adpositions. In other words, we are dealing with two morphosyntactic case values, however they are realized, and not with morphological overabundance (Thornton 2019). We do not propose additional case values in such instances (see Corbett 2012: 215–216 for discussion, following Zaliznjak 1973: 62). The slash indicates only that there is an alternation of case values. The conditions on this alternation can be of different types, not just semantic as discussed in this section (see further in §5). The relative frequency of A and B may be balanced or skewed.

Table 10 represents a relatively common situation, which was previewed in the Russian data in **Table 4** (there involving locative vs accusative). Here we illustrate from German. We find some prepositions which govern the dative (like *mit* ‘with’), and others which govern the accusative (like *durch* ‘through’). Those which interest us can govern both dative and accusative; they include *auf* ‘on(to)’, *hinter* ‘behind’, *in* ‘in(to)’ and *neben* ‘near’ (Draye 2014). As the glosses suggest, motion is involved in the distinction:

- (7) *German* (deu, Draye 2014: 95)
 Sein-e Brille lieg-t auf **dem** / *den Tisch.
 3SG.M.POSS-SG.F glasses lie-3SG on DET.SG.M.DAT / DET.SG.M.ACC table
 ‘His glasses are lying on the table.’
- (8) Er hat sein-e Brille auf **den** / *den Tisch
 3SG.M have.3SG 3SG.M.POSS-SG.F glasses on DET.SG.M.ACC / DET.SG.M.DAT table
 gelegt
 put.PST.PTCP
 ‘He put his glasses¹³ on(to) the table.’
 (Draye 2014: 95)

Broadly speaking, with the adpositions which allow this alternation, the dative (7) indicates location (place), while the accusative as in (8) is used for motion (path). And while many instances are clear-cut following this criterion, there is much more to be said about the less

¹³ *Brille* ‘glasses’ is a normal singular noun in German, not a plurale tantum noun.

obvious examples (Smith 1995; Draye 2014). The German evidence involving just these case values (omitting nominative and genitive) is summarized in **Table 11**.

	governee₁	governee₂	governee₃
<i>mit</i> ‘with’	DATIVE	DATIVE	DATIVE
<i>auf</i> ‘on(to)’	DATIVE / ACCUSATIVE	DATIVE / ACCUSATIVE	DATIVE / ACCUSATIVE
<i>durch</i> ‘through’	ACCUSATIVE	ACCUSATIVE	ACCUSATIVE

Table 11: Split adpositions: plus alternation of case values (German).

The basic semantic distinction of location (place) versus motion (path) is widely found (see, for example, Lestrade 2006a: 89 on Russian); in many accounts they are assigned different structures (Garzonio & Rossi 2020a: 7–9 and references there).¹⁴ Note, however, that the strategy above, where the expression of motion depends just on the case value, is common in Indo-European languages, but infrequent beyond these languages (Lestrade & de Schepper & Zwarts 2011: 266). For the related directive-locative alternation see Zaika (2016). Here, and more generally, we need to spell out the conditions on “/”; that is, we need to specify the factors that determine each case value. The condition may be clear-cut, as with examples (7) and (8). However, there are other instances where adpositions govern different case values and possible motivating factors are hard to establish. And at the extreme, the alternatives may be so different as to raise the issue of different lexical entries. Thus **Table 5** includes the Russian preposition *o*; when taking the accusative it means ‘against’ (as in a collision), while when taking the locative it means ‘about, concerning’. It is debatable whether this should be treated as one adposition (with a split in case government) or two adpositions.

More generally, we have a split in government provided that: (i) we have a reasonable assumption that the value will be the same; but (ii) it is different. And indeed, given the numerous canonical situations where there is a single case value, it is reasonable to enquire further about those instances where specifying both the adposition and the governee is not sufficient to give the case value that will be found. We should not ‘hard wire’ our expectations, by assuming that the condition will be of a particular nature, for instance, semantic. This is one type of possible condition, but just one in the list. Besides being of different types, conditions generalize over the different situations we are discussing. Hence we reserve discussion of them for §5.

4.2 Splits of governees with alternation of case values

We now switch to the other dimension, where the alternation depends on the governee, as in **Table 12**.

¹⁴ For a broader typology of spatial relations, see Nintemann & Vorholt & Arbes & Behrens (2025).

	governee₁	governee₂	governee₃
adposition ₁	A	A/B	B
adposition ₂	A	A/B	B
adposition ₃	A	A/B	B

Table 12: Split governees: plus alternation of case values (schematic).

In this system, the governee determines the outcome of the case alternation. A possible example might have these characteristics: governee₁ could be the first and second person pronouns, taking one case value; governee₃ could be nouns, taking another; and governee₂ could be the third person pronoun, allowing an alternation between both case values. I have not encountered this system in pure form. However, a comparable situation does occur as a part of more complex systems. Thus in Modern Eastern Armenian, many adpositions take the dative of nouns, as in (9):

- (9) *Modern Eastern Armenian* (hye, Dum-Tragut 2009: 300)
 im **ėnker-oj** het mišt hayeren em xos-um.
 1SG.GEN friend-DAT with always Armenian AUX.1SG speak-PTCP.PRS
 ‘I always speak Armenian with my friend.’

Note that for nouns there is no distinct form of the genitive; the forms traditionally called dative function as both genitive and dative. Nouns are included for completeness only; it is the pronouns which are our focus here, since they are the locus of the split. Pronouns have separate forms for the two case values. In written and standard Armenian, we find adpositions governing third person personal pronouns in the genitive:

- (10) **nra** het mišt hayeren em xos-um.
 3SG.GEN with always Armenian AUX.1SG speak-PTCP.PRS
 ‘I always speak Armenian with him.’
 (Dum-Tragut 2009: 300)

For the first and second persons, however, in written and standard Armenian, we find the dative:

- (11) **inj** het mišt hayeren ē xos-um.
 1SG.DAT with always Armenian AUX.3SG speak-PTCP.PRS
 ‘He always speaks Armenian with me.’
 (Dum-Tragut 2009: 300)

In colloquial Armenian, the genitive (found with third person pronouns, and demonstratives) is generalized to the first and second persons:

- (12) **im** het mišt hayeren ē xos-um.
 1SG.GEN with always Armenian AUX.3SG speak-PTCP.PRS
 ‘He always speaks Armenian with me.’
 (Dum-Tragut 2009: 300)

For the sake of completeness we note that in colloquial Armenian the personal pronoun may also be replaced here by the possessive suffix:

- (13) het-s mišt hayeren ē xos-um.
 with-1SG always Armenian AUX.3SG speak-PTCP.PRS
 ‘He always speaks Armenian with me.’
 (Dum-Tragut 2009: 300)

The important point is that for a substantial part of the adpositional system, though certainly not for all, there is an alternation of case values which is determined by the governee, specifically the particular pronoun involved, as represented in **Table 13** (again nouns are included only for completeness).¹⁵

	personal pronoun 3	personal pronoun 1,2	noun
adposition ₁	GENITIVE	GENITIVE/DATIVE	DATIVE
adposition ₂	GENITIVE	GENITIVE/DATIVE	DATIVE
adposition ₃	GENITIVE	GENITIVE/DATIVE	DATIVE

Table 13: Split governees: plus alternation of case values (Modern Eastern Armenian).

The situation in Modern Eastern Armenian does indeed show partitioning of the governees and an alternation of case values. It is somewhat marginal in that it affects only the pronouns. And this raises the issue which often arises with pronouns, namely whether the split is to be specified in featural terms (according to person) or whether the pronouns should be treated as different lexemes (Corbett 2005).

4.3 Partitioning of adpositions and governees with alternations of case values

We now move to the situation, diagrammed in **Table 14**, where there is a split in the adpositions, and a split in the governees, both contributing to determining the outcome of an alternation of case values.

	governee ₁	governee ₂	governee ₃
adposition ₁	A	A	A
adposition ₂	A	A/B	B
adposition ₃	B	B	B

Table 14: Split adpositions, split governees, plus alternation of case values (schematic).

¹⁵ Note that the label we attach to the noun forms does not affect the need to recognize a split; the pronouns distinguish genitive from dative, and the choice depends in part on the particular pronoun.

This might seem unlikely, but we have a carefully described instance of just this situation. It is found in Finnish, as analysed by Lestrade (2010). Finnish has some adpositions which govern the genitive (the more common situation) and some which govern the partitive (see Huumo 2024 for rich data on the partitive). There are four adpositions which govern both genitive and partitive: *lähe-* ‘near’, *keske-* ‘in the middle’, *ympäri* ‘(a)round’, and *yli* ‘over’,¹⁶ as illustrated in (14) and (15):

- (14) Finnish (fin, Lestrade 2010: 604)
 Juoks-i-mme **kaupungi-n** ympäri.
 run-PST-1PL city-GEN around
 ‘We ran around the city.’
- (15) Juoks-i-mme ympäri **kaupunki-a**.
 run-PST-1PL around city-PART
 ‘We were running around in the city.’
 (Lestrade 2010: 604)

These examples illustrate two general regularities. First, these adpositions occur as both postpositions and prepositions;¹⁷ see Huumo (2013) for discussion. As postpositions they more often assign genitive case as in (14), while as prepositions they normally assign partitive case (15); however, this is only a tendency (Lestrade 2006b: 27). Second, we see a more literal meaning in (14) and a more abstract meaning in (15). There are additional complications; for instance, within the partitive case, combining with a preposition gives the more literal meaning and combining with a postposition yields a more abstract meaning (Lestrade 2010: 605). The main point for us is that the adposition plays a key role in the split; mainly in that it is only some adpositions that induce the choice, and secondly in that the position (preposition or postposition) also has an effect.

What then of the other dimension? Does the governee matter? It turns out that it does. Lestrade (2010) conducted a corpus investigation of five newspapers (approximately 27 million words) and checked *lähe-* ‘near’, *keske-* ‘in the middle’ and *ympäri* ‘(a)round’. Objects appearing fewer than three times were excluded, and this yielded 3671 examples (Lestrade 2010: 618). While the three adpositions should be able to assign either case to a large number of nouns, in fact for any given noun they typically assign the same case. Just 25 nouns are found in both genitive and partitive with the same adposition (Lestrade 2010: 623); information on all 25 is

¹⁶ The hyphen indicates that these adpositions take case affixes; for the contribution of this type of case marking, in Finnish and more generally, see Lestrade & de Schepper & Zwarts (2011); for the wider Uralic perspective see Grünthal (2022). The adposition *yli* ‘over’ is special in that the alternation is found only in non-spatial use (see Lestrade 2010: 626fn1), and was not included in the corpus counts. For detailed and comparative information on *yli* ‘over’ in temporal use, see Björklund & Viimaranta (2019).

¹⁷ They may be called ‘ambipositions’; see Libert (2006: 1) for discussion of the term. Huumo & Lehismets (2011) offer analysis of the meaning differences between prepositional and postpositional use with adpositions indicating a path.

given in **Table 15**. The axes have been switched compared with **Table 14** and similar tables; those combinations of adposition and governee where both case values are found are again indicated with a heavy frame. To be clear, for a noun to be included requires that there is at least one adposition which takes this noun in both genitive and partitive in the corpus.

governee	adposition		
	<i>lähe-</i> ‘near’	<i>keske-</i> ‘in the middle of’	<i>ympäri</i> ‘around’
<i>asutukse</i> ‘settlement’	GEN/PART	GEN/PART	–
<i>elämä</i> ‘life’	–	GEN/PART	–
<i>huonee</i> ‘room’	–	GEN/PART	PART
<i>ihminen</i> (PL) ‘human, man’ ¹⁸	GEN/PART	GEN	–
<i>kaupunki</i> ‘city’	GEN/PART	GEN/PART	PART
<i>kenttä</i> ‘field, area’	PART	PART	GEN/PART
<i>keskusta</i> ‘centre’	GEN/PART	–	PART
<i>kirkko</i> ‘church’	GEN/PART	GEN	GEN
<i>koti</i> ‘home’	GEN/PART	–	–
<i>koulu</i> ‘school’	GEN	GEN	GEN/PART
<i>kylä</i> ‘village’	GEN	GEN/PART	PART
<i>lentokenttä</i> ‘airport’	GEN/PART	–	–
<i>luonto</i> ‘nature, wildlife’	PART	GEN/PART	–
<i>maa</i> ‘country, earth’	GEN/PART	GEN	PART
<i>maailma</i> ‘world’	GEN	–	GEN/PART
<i>maapallo</i> ‘globe’	–	–	GEN/PART
<i>metsä</i> ‘forest, woods’	–	GEN/PART	–
<i>pääkaupunki</i> ‘capital’	GEN/PART	–	–
<i>raja</i> ‘boundary, limit’	GEN/PART	–	–
<i>ranta</i> ‘border, edge’	GEN/PART	–	–
<i>rautatieasema</i> ‘railway station’	GEN/PART	–	–
<i>sota</i> ‘war’	–	GEN/PART	–
<i>talo</i> ‘house’	GEN	PART	GEN/PART
<i>tori</i> ‘marketplace’	–	GEN/PART	GEN
<i>vuosi</i> ‘year’	–	GEN/PART	GEN

Table 15: Split adpositions, split governees, plus alternation of case values (Finnish).

¹⁸ The restriction to plural is for semantic compatibility; the singular would give the ‘inside’ reading.

To appreciate **Table 15**, it is important to recall that there are adpositions which govern a single case value: *sisä* ‘in’ takes the genitive, while *päin* ‘against’ takes the partitive (Lestrade 2006b: 24). These would form an outer frame to **Table 15**. This table homes in on those adpositions which in principle take both genitive and partitive (based on Lestrade 2010: 623). In his sizeable corpus, only a small number of nouns appeared in both possible cases, and then not with all combinations of possible adpositions and possible nouns. Lestrade’s example noun *kaupunki* ‘city’ as in (14) and (15) is in a sense the “best” example. It is found with both cases in the corpus with *lähe*- ‘near’ and *keske*- ‘in the middle’; not, however, with *ympäri* ‘around’, where only the partitive appeared, even though both case values are in principle possible. On the other hand, five other nouns do appear with both possible case values when governed by *ympäri* ‘around’.

Table 15 demonstrates that the case value depends on both adposition and governee. Moreover, the environments where an alternation is found are quite specific: particular combinations of adposition and noun governee. Only a small number of nouns (25) allow the alternation. And indeed no noun alternates with all three adpositions. Several combinations are not attested in the corpus (indicated ‘-’). Finnish thus provides a fine example of the situation diagrammed in **Table 14**, and Lestrade’s investigation allows us to appreciate the detail.

The role of frequency is interesting here. The 25 nouns which show the case alternation have a significantly higher frequency than nouns which consistently take one case value.¹⁹ Note, however, that there are some highly frequent nouns which do not take both cases (Lestrade 2010: 623). Some of the gaps in **Table 14** are inherent (Chuang & Brown & Baayen & Evans 2022: 396);

¹⁹ Here are Lestrade’s data:

Group	n types	n tokens	Mean (log) frequency
Genitive	175	1241	7.1 (0.85)
Partitive	64	561	8.8 (0.94)
Different	8	132	16.5 (1.2)
Alternating	25	1737	69.5 (1.8)
Total	272	3671	Overall mean (log): 13.5 (1.13)

Table i: Case distribution per group (Lestrade 2010: 621).

Nouns are split into four groups: those which appeared in the genitive only, or in the partitive only; those that occurred in different cases, but always in the same case-adposition combinations (a noun might occur only in the genitive case for *keske*- ‘in the middle’ and only in the partitive with *lähe*- ‘near’); and finally the group that interests us most, the alternating group, the nouns for which the same adposition(s) govern different cases. The frequency of these case alternating nouns is significantly higher than that of nouns that are consistently assigned one particular case: Wilcoxon rank sum test $W = 5772$, p -value ~ 0 (Lestrade 2010: 622–623). For data on the overall frequencies of case values in Finnish see Räsänen (1979).

even with a much larger corpus we would not expect to find examples. Others are contingent; given sufficient data we might well come across an example. (See also Nikolaev & Bermel 2023 for discussion of missing forms in corpora.) For our purposes, the important examples are those in **Table 14** where we do find alternation. And to identify those (and indeed to investigate further the missing cells) we need to refer both to the governor (the adposition) *and* to the governee (the noun). Now certain other alternations discussed earlier have been described in terms of an opposition between pronoun and noun as governees. It seemed surprising that the part of speech should matter, since it implies that the adposition “peeks” ahead when determining its case government. But that is what we also find in the partitioning of the noun inventory. The two issues can potentially be brought together, with frequency playing a role, since we expect pronouns to be more frequent than individual nouns.

A further instance of this general type, an alternation involving both the adposition and the governee, can be found in Russian, involving the single preposition *po* ‘for’ (also ‘on, about, by, according to’); there is a detailed account in Muravenko (2014), discussed in Corbett (2023a: 129–130), together with a range of other sources, and see also Severskaia (2022). The historical development of this alternation shows that both types of governee restriction were involved (§3.2); both the part of speech of the governee (noun vs pronoun) and its feature value (singular vs plural) played a role. For the analogous situation in Belarusian see Corbett (2023a: 126–127).

4.4 Aside: partitioning of adpositions and governees with alternations of functional adpositions

We have been considering the partitioning of adpositions and governees with alternations of case values. An intriguing comparison is found in Italian (Garzonio & Rossi 2020b). It requires us to recognize a distinction between lexical prepositions (like *sotto* ‘under’) and ‘functional’ prepositions (*a* ‘to’ or *di* ‘of’). In complex prepositional phrases, a lexical preposition may take a functional preposition, as in (16):

- (16) *Italian* (ita, Garzonio & Rossi 2020b: 85)
- | | | | |
|-----------------|-----|-----------|-------------|
| sotto | (a) | un | tavol-o |
| under | to | INDF.SG.M | table(M)-SG |
| ‘under a table’ | | | |

Whether a functional preposition is found or not depends *both* on the lexical preposition *and* on the governee (complement). The lexical prepositions are of three types and the governees can be divided into full nominal governees or pronouns. The possibilities are schematized in **Table 16** (based on Garzonio & Rossi 2020b: 88). Note that there is considerable inter-speaker variation.

group of preposition	type of complement	
	full nominal complement	pronominal complement
Group 1 e.g. <i>accanto</i> ‘beside’	<i>a</i>	<i>a</i>
Group 2 e.g. <i>sotto</i> ‘under’	<i>a/∅</i>	<i>a/di</i>
Group 3 e.g. <i>verso</i> ‘toward’	$\emptyset$	<i>di</i>

Table 16: Alternation in functional prepositions according to the lexical preposition and the complement (Italian).

It is striking that to determine whether or not there will be a functional preposition (and which it will be) we need to partition the lexical prepositions, and of the governees. This double source of determination is analogous to what we see with case government in Finnish.

5 Conditions on alternations

We started from the simple canonical situation, where all constructions involving an adposition and a governee would have the same case value. That situation is rare. We then investigated splits, and found many examples. These may involve the adpositions, the governees, or both (§3). We then analysed instances where even allowing for splits in the adposition, the governee or both, we had to invoke further information to account for alternations (§4). Here I consider this further information, the ‘conditions’ on the alternations. I take two perspectives: the type of information (§5.1) and the locus of this information (§5.2).

5.1 Conditions on alternations: types of information

We have already seen examples where the condition on an alternation is **semantic**. The widespread distinction between location and motion, seen for German in §4.1, is clearly semantic. We also find **syntactic** conditions. A notable one is word order, where the relative order of adposition and governee plays a role. For example, Draye (2014: 120–121fn2) states that German *entlang* ‘along’ has different possibilities according to whether it behaves as a preposition or postposition: as a preposition it takes accusative, genitive and dative, but as a postposition accusative and dative only.²⁰ And in Finnish (§4.3) we noted the strong link between the order of adposition and governee and the case value selected.

²⁰ There are instances where the preposed governee has a dependent genitive which then appears directly before postposed *entlang* ‘along’.

A further type of condition is register, which is related to the issue of **diachrony**. As the case governed by adpositions changes, conservative usage is likely to be perceived as stylistically marked. Thus in Russian, at the beginning of the 19th century, *meždu* ‘between, among’ could take genitive or instrumental. The instrumental was the more frequent, and it became increasingly dominant through the 19th century (Filippova 1964: 79–83). In the 20th century the genitive was still found, with Graudina, Ickovič & Katlinskaja (1976: 57) suggesting that it tends to occur in poetical usage (see also Kiričenko 2000 on its semantics). In German, there has been movement in adpositional government, for particular subsets of adpositions, both towards the dative *and* towards the genitive, interacting with register, as documented by Di Meola (2004); and most recently see Vieregge (2025) on speakers’ reactions to the alternation. In Armenian too (§4.2), we noted change in the system leading to differences in register of the possible case outcomes.

5.2 Conditions on alternations: locus of information

I now ask where such information should be located; in particular, does it logically attach to the lexical entries of adpositions (§4.1), is it part of the specification of governees (§4.2) or both (§4.3)? The picture is not always clear-cut, but in most instances the information primarily adds to the lexical entries of adpositions. In the semantic condition location vs motion, this is a generalization over a subset of the adpositions. (Note that we cannot assign the difference directly to the case values, since these typically have varied semantics going well beyond location and motion; furthermore it is only a subset of the potentially relevant adpositions which are involved, and so these have to be specified.) On the syntactic condition involving word order, the evidence again points to locating this with the lexical entry for particular adpositions. And for the examples cited of conditions relating to register, it appears again that the lexical entries of the individual adpositions are the logical place for the additional conditions on alternations. Further research may well establish conditions which are located on the governee, or on both adposition and governee.

6 The alternative perspective: Differential marking

I started from a simple, deterministic system, and gradually relaxed the constraints, allowing us to specify more complex systems. I now ask what happens if we take the opposite perspective. I try the analysis starting from the situation where all case values are freely available to all adpositions for all governees, as represented in **Table 17**.

	governee₁	governee₂	governee₃
adposition ₁	A/B/C	A/B/C	A/B/C
adposition ₂	A/B/C	A/B/C	A/B/C
adposition ₃	A/B/C	A/B/C	A/B/C

Table 17: All values possible for all combinations (differential approach, schematic).

In the situation given in **Table 17**, all adpositions can take all case values for all governees (the opposite of **Table 1**). This should be considered as a potential baseline for a typology. We need to ask whether it can be constrained in principled ways, in order to describe the systems we actually find. For each restriction we would need to specify which items are involved, and what the condition is.

We can envisage different types of restriction. First, **semantic** conditions. It could be that some values would be more likely with some adpositions than others, for semantic reasons; in a given language this might be a hard constraint. For instance, accusative case might be associated with motion, so adpositions indicating direction take the accusative, which may be a general regularity (perhaps with some exceptions and attendant splits). There could also be **syntactic** constraints, limiting the degree of flux seen in **Table 17**. We noted in §5 that the case values governed by Finnish adpositions are strongly influenced by word order. As we build in more restrictions, whether semantic or syntactic, our descriptions become increasingly like the ones we proposed earlier. In some instances, these constraints would be quite severe; looking back to the Russian system in **Table 5**, we would need to ensure that *čerez* ‘across’ could *not* take the genitive, dative, instrumental, or locative cases (the nominative would be ruled out by a more general constraint).

Hence in principle we could have a schema as in **Table 17** as our baseline (all values possible for all combinations) or the one we adopted earlier, as in **Table 1** (all adpositions govern the same unique feature value of every governee). While both approaches are possible, there are two main considerations which favour the one we have taken (as in **Table 1**). First, starting from a single fixed reference point is in principle preferable to working from a baseline where everything varies; this is because it is easy to be sure of the single point of reference and quite hard to be sure that one has all the potential types of variability under control. And second, in practical terms, the real systems we wish to calibrate are much closer to the fixed reference point. That is, we find systems which are largely fixed, but with smaller parts (subsystems) which show variability, and would be suited to a differential approach. We do not find systems which are through-and-through differential, nor systems that are largely differential but with pockets of fixed government. Rather we find differential subsystems within otherwise fixed systems.

This reasoning scales up, arguably, to differential analyses more generally. That is, the systems analysed as showing differential marking do not function as whole systems, but rather are deviations within more fixed systems. Those who have pointed out the interest of differential marking, and charted it, have advanced the field considerably, and there is a large literature (see, for example, Sinnemäki 2014 for cogent discussion, and Witzlack-Makarevich & Seržant 2018 for a helpful overview). But differential subsystems are not qualitatively different; this is particularly true since the old distinction between overt vs non-overt marking in differential systems is no longer a key factor, and alternations between two overt markers are included within differential-style

analyses (Mardale & Karatsareas 2020: 1–2). Rather, differential subsystems form a part of the large typological space of marking (and particularly case-marking), which can be determinate and differential to varying degrees (compare Bresnan & Dingare & Manning 2001).

Why do we not find fully differential systems? These would imply an explosion of possibilities, which would run up against the Principle of Contrast (Clark 1987). The Finnish data (**Table 15**) are helpful here. Even in the small section of the system which allows alternatives, the forms found are a small portion of the theoretically possible: few nouns appear with both case values with the same adposition. Some are excluded by semantic incompatibility, but others, it seems, may just be highly infrequent. Since for a given adposition most nouns when governed typically stand in just one of the alternative case values, it is of little consequence which value appears.

The discussion of **Table 17** aimed to show that we could work from this end point and, by imposing conditions on such a schema, arrive at schemas like those we have been working with. The argument was a purely synchronic one. But we can also take a diachronic perspective; thus Seržant & Rafiyenko (2021: 194–197) demonstrate how in the history of Greek, adposition plus governee: (i) began as a phrase with compositional meaning (as in **Table 17**); (ii) then the different case values were used to disambiguate a particular meaning of the preposition idiosyncratically (no longer compositionally); (iii) leading to a stage where case assignment is rigid and lexicalized (closer to a canonical situation). Thus while we were concerned with the analysis of synchronic systems, we see how the different types are also related diachronically.

7 Conclusion

I have offered a canonical account of the government properties of adpositions, giving the criteria which allow us to lay out the possibilities. In particular, I separated out the determining role of the adposition, the governee, and their interaction. This typology fits within a larger typology of external splits, where government is just one of the four phenomena covered (Corbett 2023a: 113–116, 120–121). The typology offered is an exploratory one, showing the theoretical possibilities and adducing real examples, but not giving an account of their distribution. It is nevertheless clear that instances where the case value is determined by the adposition are common, others are rare (as we would expect). I went on to instances where, even given the adposition and the governee, there remains a choice in case value, and I outlined the factors that come into play. The value of the canonical approach has been to differentiate the different logical types and to highlight the interest of the rarer systems, as in Mari, Armenian and Finnish.

An important thread running through several sets of data is the role of frequency. Pronouns are likely to be frequent. And indeed, sometimes the pronouns were singled out as having special behaviour, sometimes to the point where adposition and pronoun become set expressions. But it was not only pronouns; this was clear for the example of the Russian second locative (nouns),

as in §3.3, and dramatically for Finnish (§4.3), where the interesting behaviour was found with highly frequent nouns.

The typology of adpositions proves fascinating in its own right. However, the instances where alternative case marking is available provide a helpful space in which to think about differential marking more generally. Having started from simple determinate systems as our baseline, we moved to those with larger numbers of factors influencing the choice. We saw that it would be theoretically possible to start from the other end, taking differential system as our baseline. These provide a less practical baseline, both for the issue of definition, and since fully differential systems are not found (rather we find differential subsystems). The main outcome here is that differential systems are not qualitatively different; rather they show more alternations and more conditions than the simpler systems. They can be analysed in a mutually consistent way.

Abbreviations

1, 2, 3: first, second, third persons, ACC: accusative, AUX: auxiliary, DAT: dative, DET: determiner, F: feminine, GEN: genitive, INDF: indefinite, INS: instrumental, LOC: locative, LOC2: second locative, M: masculine, NOM: nominative, PART: partitive, PL: plural, POSS: possessive, PRS: present, PST: past, PTCP : participle, SG: singular.

Acknowledgements

The support of the Leverhulme Trust, grant number RPG-2020-078 “Declining case: inflectional loss in progress” is gratefully acknowledged (<https://www.smg.surrey.ac.uk/projects/declining-case-inflectional-loss-in-progress>). I thank Peter Arkadiev, Matthew Baerman, Jules Bouton, Dávid Gyórfi, Borja Herce, Daniel Hieber, Steven Kaye, Masha Kyuseva, Alan Libert, Jérémy Pasquereau, Ilja Seržant, Kaius Sinnemäki and Anna Thornton, and particularly the *Glossa* referees, for generously providing constructive comments. I also thank Johan Rooryk for being a challenging and helpful editor. Versions were presented to the CLEAR group, UiT The Arctic University of Norway, 3 October 2024, and at ISMo 2025: 5th International Symposium of Morphology, 9 October 2025; I am grateful to both audiences for their useful suggestions.

Competing interests

The author has no competing interests to declare.

References

- Alhoniemi, Alho. 1993. *Grammatik des Tscheremissischen (Mari): mit Texten und Glossar*. Hamburg: Buske.
- Ambrasas, Vytautas & Geniušienė, Emma & Girdenis, Aleksas & Sližienė, Nijolė & Tekorienė, Dalija & Valeckienė, Adelė & Valiulytė, Elena. 1997. *Lithuanian grammar*. Vilnius: Baltos Lankos.
- Björklund, Martina & Viimaranta, Johanna. 2019. Russian, Swedish, and Finnish adpositions meaning ‘under’ and ‘over/across’ in temporal constructions. *Poljarnyj vestnik: Norwegian Journal of Slavic Studies* 22. 1–29. DOI: <https://doi.org/10.7557/6.4855>
- Blake, Barry J. 1994. *Case*. Cambridge: Cambridge University Press.
- Bond, Oliver. 2013. A base for canonical negation. In Brown, Dunstan & Chumakina, Marina & Corbett, Greville G. (eds.), *Canonical morphology and syntax*, 20–47. Oxford: Oxford University Press.
- Bond, Oliver & Chumakina, Marina & Kaye, Steven. 2023. Unusual agreement targets in unusual domains. In Chumakina, Marina & Bond, Oliver & Kaye, Steven (eds.), *Agreement beyond the verb: Unusual targets, unexpected domains*, 1–47. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oso/9780192897565.003.0001>

- Bonmann, Svenja & Riesberg, Sonja & Himmelmann, Nikolaus P. 2025. Differential object marking in Western Malayo-Polynesian symmetrical voice languages. *Linguistic Typology* 29(1). 127–165. DOI: <https://doi.org/10.1515/lingty-2024-0002>
- Bresnan, Joan & Dingare, Shipra & Manning, Christopher D. 2001. Soft constraints mirror hard constraints: Voice and person in English and Lummi. In Butt, Miriam & King, Tracy Holloway (eds.), *Proceedings of the LFG 01 Conference: The University of Hong Kong*. Stanford: CSLI Publications. <http://csli-publications.stanford.edu>
- Breu, Walter. 1994. Der Faktor Sprachkontakt in einer dynamischen Typologie des Slavischen. In Mehlig, Hans R. (ed.), *Slavistische Linguistik 1993*, 41–64. Munich: Otto Sagner.
- Brown, Dunstan. 2007. Peripheral functions and overdifferentiation: The Russian second locative. *Russian Linguistics* 31. 61–76. DOI: <https://doi.org/10.1007/s11185-006-0715-5>
- Brown, Dunstan & Hippisley, Andrew. 2012. *Network Morphology: A defaults-based theory of word structure*. Cambridge: Cambridge University Press. DOI: <https://doi.org/10.1017/CBO9780511794346>
- Burukina, Irina. 2023. On the syntax of postpositional phrases in Mari: Choosing between two structures. *Journal of Uralic Linguistics* 2(2). 158–193. DOI: <https://doi.org/10.1075/jul.00021.bur>
- Chuang, Yu-Ying & Brown, Dunstan & Baayen, R. Harald & Evans, Roger. 2022. Paradigm gaps are associated with weird “distributional semantics” properties: Russian defective nouns and their case and number paradigms. *The Mental Lexicon* 17(3). 395–421. DOI: <https://doi.org/10.1075/ml.22013.chu>
- Clark, Eve V. 1987. The principle of contrast: A constraint on language acquisition. In MacWhinney, Brian (ed.), *Mechanisms of language acquisition*, 1–33. Hillsdale, NJ: Lawrence Erlbaum.
- Corbett, Greville G. 2005. Suppletion in personal pronouns: Theory versus practice, and the place of reproducibility in typology. *Linguistic Typology* 9(1). 1–23. DOI: <https://doi.org/10.1515/lity.2005.9.1.1>
- Corbett, Greville G. 2006. *Agreement*. Cambridge: Cambridge University Press.
- Corbett, Greville G. 2010. Classic problems at the syntax-morphology interface: Whose are they? In Müller, Stefan (ed.), *Proceedings of the HPSG 2010 Conference: Université Paris Diderot, Paris 7, France, July 09–10, 2010*, 255–268. Stanford: CSLI Publications. <https://proceedings.hpsg.xyz/issue/view/35>. DOI: <https://doi.org/10.21248/hpsg.2010.14>
- Corbett, Greville G. 2012. *Features*. Cambridge: Cambridge University Press.
- Corbett, Greville G. 2023a. The typology of external splits. *Language* 99(1). 108–153. <https://muse.jhu.edu/article/884311>
- Corbett, Greville G. 2023b. The typology of external splits: Supplementary material. *Language* 99(1). *Project MUSE*. DOI: <https://doi.org/10.1353/lan.2023.0008>
- Di Meola, Claudio. 2004. The rise of the prepositional genitive in German—a grammaticalization phenomenon. *Lingua* 114. 165–283. DOI: [https://doi.org/10.1016/S0024-3841\(03\)00033-0](https://doi.org/10.1016/S0024-3841(03)00033-0)

- Draye, Luk. 2014. German two-way prepositions and related phenomena. In Delbecque, Nicole & Lahousse, Karen & Van Langendonck, Willy (eds.), *Non-nuclear cases* (Case and Grammatical Relations Across Languages 6), 95–125. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/cagral.6.04dra>
- Dryer, Matthew S. 2013. Order of adposition and noun phrase. In Dryer, Matthew S. & Haspelmath, Martin (eds.), *The World Atlas of Language Structures Online*. Leipzig: Max Planck Institute for Evolutionary Anthropology. (Available at <http://wals.info/chapter/85>, accessed 2024-02-01.)
- Dum-Tragut, Jasmine. 2009. *Armenian: Modern Eastern Armenian*. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/loall.14>
- Filippova, V. M. 1964. Izmenenija v glagol'nyx slovosocetanijax. In Vinogradov, V. V. & Švedova, N. Ju. (eds.), *Očerki po istoričeskoj grammatike russkogo literaturnogo jazyka XIX veka: IV: Izmenenija v sisteme slovosocetanij v russkom literaturnom jazyke XIX veka*, 17–127. Moscow: Nauka.
- Foley, William A. 2023. Word-class systems and other grammatical properties. In van Lier, Eva (ed.), *The Oxford handbook of word classes*, 109–133. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oxfordhb/9780198852889.013.7>
- Fortson, Benjamin W. 2010. Reconsidering the history of Latin and Sabellic adpositional morphosyntax. *The American Journal of Philology* 131(1). 121–154. <http://www.jstor.org/stable/40730915>
- Garzonio, Jacopo & Rossi, Silvia. 2020a. Variation in P: An introduction. In Garzonio, Jacopo & Rossi, Silvia (eds.), *Variation in P: Comparative approaches to adpositional phrases*, 1–16. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oso/9780190931247.003.0001>
- Garzonio, Jacopo & Rossi, Silvia. 2020b. AxParts and case in complex PPs: Microvariation in Italian dialects. In Garzonio, Jacopo & Rossi, Silvia (eds.), *Variation in P: Comparative approaches to adpositional phrases*, 84–113. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oso/9780190931247.003.0004>
- Goddard, Cliff. 1982. Case systems and case marking in Australian languages: A new interpretation. *Australian Journal of Linguistics* 2(2). 167–196. DOI: <https://doi.org/10.1080/07268608208599290>
- Gouskova, Maria. 2019. Phonological words in the syntax and in the lexicon: A study of Russian prepositions. *Journal of Slavic Linguistics* 27(2). 161–212. <https://www.jstor.org/stable/27095644>
- Graudina, L. K. & Ickovič, V. A. & Katlinskaja, L. P. 1976. *Grammatičeskaja pravil'nost' russkoj reči: opyt častotno-stilističeskogo slovarja variantov*. Moscow: Nauka.
- Grünthal, Riho. 2019. Canonical and non-canonical patterns in the adpositional phrase of Western Uralic: Constraints of borrowing. *Suomalais-Ugrilaisen Seuran Aikakauskirja / Journal de la Société Finno-Ougrienne* 97. 9–34. DOI: <https://doi.org/10.33340/susa.70211>
- Grünthal, Riho. 2022. Adpositions and adpositional phrases. In Bakró-Nagy, Marianne & Laakso, Johanna & Skribnik, Elena & Grünthal, Riho (eds.), *The Oxford guide to the Uralic languages*, 961–969. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oso/9780198767664.003.0050>
- Hagège, Claude. 2010. *Adpositions*. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/acprof:oso/9780199575008.001.0001>

- Hentschel, Gerd. 2003. Zur Klassifikation von Präpositionen im Vergleich zur Klassifikation von Kasus. In Hentschel, Gerd & Menzel, Thomas (eds.), *Präpositionen im Polnischen*, (Studia Slavica Oldenburgensia 11), 161–191. Oldenburg: Bibliotheks- und Informationssystem der Carl von Ossietzky Universität Oldenburg.
- Herce, Borja. 2023. Adpositions. In van Lier, Eva (ed.), *Oxford handbook of word classes*, 420–442. Oxford: Oxford University Press. DOI: <https://doi.org/10.1093/oxfordhb/9780198852889.013.12>
- Hill, Steven P. 1977. *The N-factor and Russian prepositions: Their development in 11th–20th century texts*. The Hague: Mouton. DOI: <https://doi.org/10.1515/9783110801910>
- Huddleston, Rodney & Pullum, Geoffrey K. (eds.). 2002. *The Cambridge grammar of the English language*. Cambridge: Cambridge University Press. DOI: <https://doi.org/10.1017/9781316423530>
- Hutchinson, John G. 2025. On prepositions and case in Scottish Gaelic. *Word Structure* 18(1–2). 1–40. DOI: <https://doi.org/10.3366/word.2025.0245>
- Huumo, Tuomas 2013. Many ways of moving along a path: What distinguishes prepositional and postpositional uses of Finnish path adpositions. *Lingua* 133. 319–335. DOI: <https://doi.org/10.1016/j.lingua.2013.05.006>
- Huumo, Tuomas. 2024. Why partitive? Possible motivations for the partitive complement of Finnic adpositions. *Eesti Ja Soome-Ugri Keeleteaduse Ajakiri / Journal of Estonian and Finno-Ugric Linguistics* 15(1). 55–100. DOI: <https://doi.org/10.12697/jeful.2024.15.1.02>
- Huumo, Tuomas & Lehimets, Kersten. 2011. Finnish path adpositions: Prepositions or postpositions, subjective or objective motion? In Bierwiazzonek, Bogusław & Cetnarowska, Bożena & Turula, Anna (eds.), *Syntax in cognitive grammar*, 135–148. Częstochowa: Wydawnictwo Wyższej Szkoły Lingwistycznej.
- Kangasmaa-Minn, Eeva. 1998. Mari. In Abondolo, Daniel (ed.), *The Uralic languages*, 219–248. London: Routledge.
- Kiričenko, A. S. 2000. Sistemnye semantičeskie xarakteristiki i oblast' denotacii predloga *među*. In Pajar, D. & Seliverstova, O. N. (eds.), *Issledovanija po semantike predlogov: Sbornik statej*, 338–352. Moscow: Russkie slovari.
- Krasovitsky, Alexander & Kyuseva, Maria & Baerman, Matthew & Corbett, Greville G. 2026. Diachronic paths to uninflectedness in South Slavic. Palancar, Enrique & Fedden, Sebastian (eds.), *Voids in morphology: Exploring “uninflectedness”*, 151–189. Berlin: Language Science Press.
- Lamb, William. 2025. *Scottish Gaelic: A comprehensive grammar*. London: Routledge. DOI: <https://doi.org/10.4324/9780429199196>
- Lestrade, Sander A. M. 2006a. *Adpositional case*. MA thesis, Linguistics Department, Radboud University Nijmegen.
- Lestrade, Sander A. M. 2006b. Marked adpositions. *Proceedings of the Third ACL-SIGSEM Workshop on Prepositions*, 23–28. Trento: Association for Computational Linguistics. DOI: <https://doi.org/10.3115/1621431.1621435>
- Lestrade, Sander. 2010. Finnish case alternating adpositions: A corpus study. *Linguistics* 48(3). 603–628. DOI: <https://doi.org/10.1515/ling.2010.019>

- Lestrade, Sander & de Schepper, Kees & Zwarts, Joost. 2011. The distribution of labor between adpositions and case within complex spatial PPs. *STUF – Language Typology and Universals* 64(3). 256–274. DOI: <https://doi.org/10.1524/stuf.2011.0018>
- Libert, Alan Reed. 2002. On the range and variety of cases assigned by adpositions. In Amberber, Mengistu & Collins, Peter (eds.), *Language universals and variation*, 131–154. Westport, Conn.: Praeger.
- Libert, Alan [Reed]. 2006. *Ambipositions*. Munich: Lincom Europa
- Libert, Alan Reed. 2013. *Adpositions and other parts of speech*. Frankfurt: Peter Lang. DOI: <https://doi.org/10.3726/978-3-653-03682-4>
- Libert, Alan [Reed]. 2016. Adpositions. *Oxford bibliographies*. DOI: <https://doi.org/10.1093/obo/9780199772810-0111>
- Ljaševskaja, Ol'ga N. & Šarov, Sergej A. 2009. *Častotnyj slovar' sovremennogo russkogo jazyka: na materialax nacional'nogo korpusa russkogo jazyka*. Moscow: Azbukovnik.
- Mardale, Alexandru & Karatsareas, Petros 2020. Differential Object Marking and Language Contact: An Introduction to this Special Issue. *Journal of Language Contact* 13. 1–16. DOI: <https://doi.org/10.1163/19552629-bja10001>
- Meľčuk, Igor A. 1985. *Poverxnostnyj sintaksis russkix čislovyx vyraženij* (= Wiener Slawistischer Almanach, Sonderband 16). Vienna: Institut für Slawistik der Universität Wien.
- Merlan, Francesca C. 1994. *A Grammar of Wardaman, a language of the Northern Territory of Australia* (Mouton Grammar Library 11). Berlin: Mouton de Gruyter. DOI: <https://doi.org/10.1515/9783110871371>
- Micali, Irene & Vincent, Nigel & Wheeler, Max W. 2025. Morphological splits: Constructions with GO + infinitive in Romance. In Wild, Mario & Negrinelli, Stefano & Breimaier, Federica & Cristelli, Stefano & Loporcaro, Michele & Paciaroni, Tania (eds.), *Italian dialects and language contact: Theory, typology, data*. Special issue of *Isogloss. Open Journal of Romance Linguistics* 11(6)/3. 1–17. DOI: <https://doi.org/10.5565/rev/isogloss.521>
- Muravenko, Elena V. 2014. Izmenenie upravlenija predloga po v russkom jazyke. *Acta Linguistica Petropolitana: Trudy instituta lingvističeskix issledovanij* 2. 643–672. Online: <https://cyberleninka.ru/article/v/izmenenie-upravleniya-predloga-po-v-russkom-yazyke>
- Ngo, Binh. 2021. *Vietnamese: An essential grammar*. London: Routledge.
- Nikolaev, Alexandre & Bermel, Neil. 2023. Studying negative evidence in Finnish language corpora. *Word Structure* 16(2–3). 206–232. DOI: <https://doi.org/10.3366/word.2023.0229>
- Nintemann, Julia & Vorholt, Maike & Arbes, Deborah & Behrens, Kevin. 2025. OUT OF our minds and ONTO the paper: On the morphological complexity of configurational spatial relations. In Levkovych, Nataliya & Nintemann, Julia & Vorholt, Maike (eds.), *Exploring structures in languages and language contact*, 147–198. Berlin: De Gruyter Mouton. DOI: <https://doi.org/10.1515/9783111496436-007>
- Noyer, Rolf. 2005. A constraint on interclass syncretism. In Booij, Geert & Marle, Jaap van (eds.), *Yearbook of morphology 2004*, 273–315. Dordrecht: Springer. DOI: https://doi.org/10.1007/1-4020-2900-4_9

- Plungian [Plungjan], Vladimir A. 2002. K semantike ruskogo lokativa. *Semiotika i informatika* 37. 229–254.
- Räsänen, Seppo. 1979. Huomioita suomen sijojen frekvensseistä [Observations on the frequency of the Finnish case forms]. *Sananjalka* 21. 17–34. DOI: <https://doi.org/10.30673/sja.86427>
- Riese, Timothy & Bradley, Jeremy & Yefremova, Tatiana. 2022. *Mari (марий үйлме): An essential grammar for international learners*. [Version 1.0] Vienna: University of Vienna. [published online at grammar.mari-language.com]
- Round, Erich R. & Corbett, Greville G. 2020. Comparability and measurement in typological science: The bright future for linguistics. *Linguistic Typology* 24(3). 489–525. DOI: <https://doi.org/10.1515/lingty-2020-2060>
- Scorza, David. 1985. A sketch of Au morphology and syntax. *Papers in New Guinea Linguistics* No. 22, 215–273. Canberra: Pacific Linguistics A63. <https://hdl.handle.net/11858/00-001M-0000-0011-54E1-1>
- Severskaia, Ol'ga. 2022. *Po priezdu, po okončanju ... Kogda načalos' grammatičeskoe var'irovanie?* In Warditz, Vladislava (ed.) *Russian Grammar: System – Usus – Variation / Russkaja grammatika: Sistema– uzus – var'irovanie*, 461–468, Berlin: Peter Lang.
- Sinnemäki, Kaius. 2014. A typological perspective on Differential Object Marking. *Linguistics* 52(2). 281–313. DOI: <https://doi.org/10.1515/ling-2013-0063>
- Smith, Michael B. 1995. Semantic motivation vs. arbitrariness in grammar: Toward a more general account of the DAT/ACC contrast with German two-way prepositions. In Rauch, Irmengard & Carr, Gerald F. (eds.), *Insights in Germanic linguistics I: Methodology in transition*, 293–323. Berlin: Mouton de Gruyter. DOI: <https://doi.org/10.1515/9783110810868.293>
- Spencer, Andrew J. & Stump, Gregory T. 2013. Hungarian pronominal case and the dichotomy of content and form in inflectional morphology. *Natural Language and Linguistic Theory* 31. 1207–1248. DOI: <https://doi.org/10.1007/s11049-013-9207-7>
- Stewart, Thomas W. & Joseph, Brian D. 2009. How big can case systems get? Evidence from Scottish Gaelic. *Word Structure* 2(1). 108–120. DOI: <https://doi.org/10.3366/E1750124509000312>
- Svenonius, Peter. 2007. Adpositions, particles and the arguments they introduce. In Reuland, Eric J. & Bhattacharya, Tanmoy & Spathas, Giorgos (eds.), *Argument structure* (Linguistik Aktuell / Linguistics Today 108), 63–103. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/la.108.08sve>
- Thornton, Anna M. 2019. Overabundance: A Canonical Typology. In Rainer, Franz & Gardani, Francesco & Dressler, Wolfgang U. & Luschützky, Hans Christian (eds.), *Competition in inflection and word-formation* (Studies in Morphology 5), 223–258. Cham: Springer. DOI: https://doi.org/10.1007/978-3-030-02550-2_9
- Timberlake, Alan. 2004. *A reference grammar of Russian*. Cambridge: Cambridge University Press.
- Vierегge, Annika. 2025. *Bewertung und Variation der Präpositionalkasus im Deutschen: Der Einfluss metapragmatischer Urteile auf die Rektion von Präpositionen*. (Open Germanic Linguistics 10). Berlin: Language Science Press. DOI: <https://doi.org/10.5281/zenodo.14608443>

Wade, Terence. 1983. *Prepositions in Modern Russian* (Durham Modern Languages Series). Durham: University of Durham.

Wade, Terence. 1992. *A comprehensive Russian grammar*. Oxford: Blackwell.

Witzlack-Makarevich, Alena & Seržant, Ilja A. 2018. Differential argument marking: Patterns of variation. In Seržant, Ilja A. & Witzlack-Makarevich, Alena (eds.), *Diachrony of differential argument marking*, 1–40. Berlin: Language Science Press. DOI: <https://doi.org/10.5281/zenodo.1228243>

Zaika, Natalia M. 2016. The directive/locative alternation in Lithuanian and elsewhere. In Holvoet, Axel & Nau, Nicole (eds.), *Argument realization in Baltic* (Valency and Grammatical Relations in Baltic 3), 333–360. Amsterdam: John Benjamins. DOI: <https://doi.org/10.1075/vargreb.3.07zai>

Zaliznjak, Andrej A. 1973. O ponimanii termina ‘padež’ v lingvističeskix opisanijax. In Zaliznjak, Andrej A. (ed.), *Problemy grammatičeskogo modelirovanija*, 53–87. Moscow: Nauka. [Reprinted in Zaliznjak, Andrej A. 2002. *Russkoe imennoe slovoizmenenie: s priloženiem izbrannyx rabot po sovremennomu russkomu jazyku i obščemu jazykoznaniju*, 613–647. Moscow: Jazyki slavjanskoj kul’tury.]

